产 品 规 格 书
Product Specification

CUSTOMER	客户：	
CUSTOMER PN	客户 PN：	
HANG CRYSTAL P/N	杭晶物料编码：	**TH53H4-40.000NQ33DTL**
MODEL	产品型号：	**TCXO SMD 5.0x3.2, CMOS, 3.3V**
NOMINAL FREQUENCY	频率：	**40.000MHz**
ISSUE DATE	日期：	**2023 / 02 / 10**

CUSTOMER'S APPROVAL

客户确认

(PLEASE RETURN A COPY WITH APPOVAL)
（请将确认的复印件返回我司）

APPROVED	QA
MB.	James J

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Revision	Description / ECN	Prepared	Approved	Date
1	Initial release	MB	James Jiang	2023-02-10
2	Not issued			
3	Not issued			
4	Not issued			

1. NOMINAL AND MAXIMUM RATINGS, OPERATING AND STORAGE CONDITIONS

	PARAMETER	SYMB.	MIN	TYP	MAX	Unit	Conditions / Remarks
1	Nominal frequency	F_N		**40.000**		MHz	–
2	Maximum supply voltage	V_{MAX}	–0.3		+4.0	V_{DC}	Between V_{CC} and GND
3	Operating supply voltage range	V_{CC}	3.135	3.30	3.465	V_{DC}	
4	Output load capacitance	CL			15	pF	
5	Operating temperature range	T_{OP}	–40	+25	+85	°C	Note 1
6	Storage Temperature Range	T_{ST}	–40		+90	°C	–

Note 1: over the whole range, the unit stays within all relevant parameter limits as specified under point 2.

2. ELECTRICAL PARAMETER LIMITS

	PARAMETER	SYMB.	MIN	TYP	MAX	Unit	Conditions / Remarks
1	Frequency tolerance	$\Delta f/F_N$	–1.0		+1.0	ppm	Offset from nominal at +25°C
2	Frequency stability	$\Delta f/F_{RFL}$	–0.5		+0.5	ppm	Note 1
3	Frequency VS voltage changes	$\Delta f/F_V$	–0.2		+0.2	ppm	Vcc ±5% at +25°C
4	Frequency VS load changes	$\Delta f/F_L$	–0.2		+0.2	ppm	
5	Aging first year	$\Delta f/F_{A1}$	–1.0		+1.0	ppm	at +25°C
6	Output voltage level HIGH	V_{OH}	2.97			V_{DC}	
7	Output voltage level LOW	V_{OL}			0.33	V_{DC}	
8	Output amplitude rise time	t_R			5.0	ns	At 20~80%V_{CC} / 15pF / +25°C
9	Output amplitude fall time	t_F			5.0	ns	At 80~20%V_{CC} / 15pF / +25°C
10	Output symmetry (Duty Cycle)	DC	45		55	%	GND level (DC cut)
11	Phase noise	L_{RMS}		-88	–80	dBc/Hz	at 10Hz offset / at +25°C
				-115	–110		at 100Hz offset / at +25°C
				-135	–130		at 1kHz offset / at +25°C
				-148	–145		at 10kHz offset / at +25°C
				-150	–145		at 100KHz offset / at +25°C
12	Startup time	t_{STRT}			5.0	ms	V_{P-P} reach >90% of amplitude
13	Current consumption	I_{CC}			10	mA	Under CL 15pF

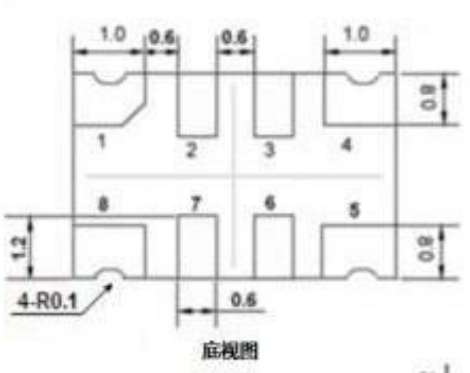
Note 1:Frequency stability is the frequency deviation over operating temperature range T_{OP} in reference to the frequency reading at +25°C.

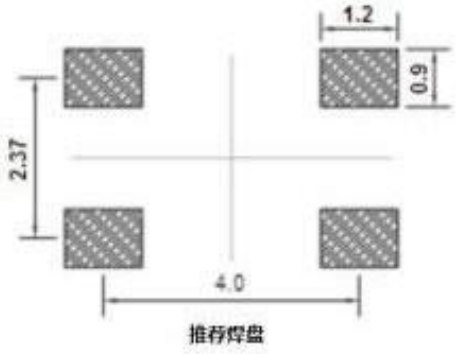
3. RODUCT MARKING

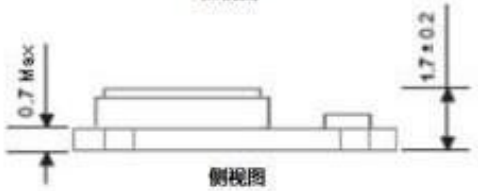
To be defined.

4. OUTLINE DRAWING

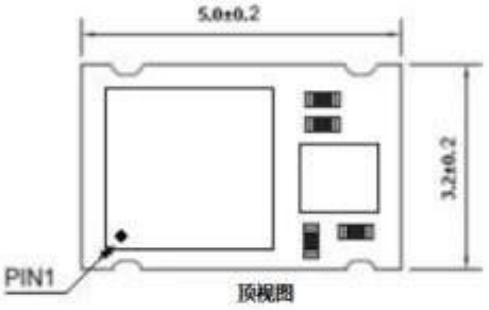
	Package description	Package model	Remarks
1	Ceramic seam seal SMD package 5.0x3.2mm with 4pads	TH53H8h17	–







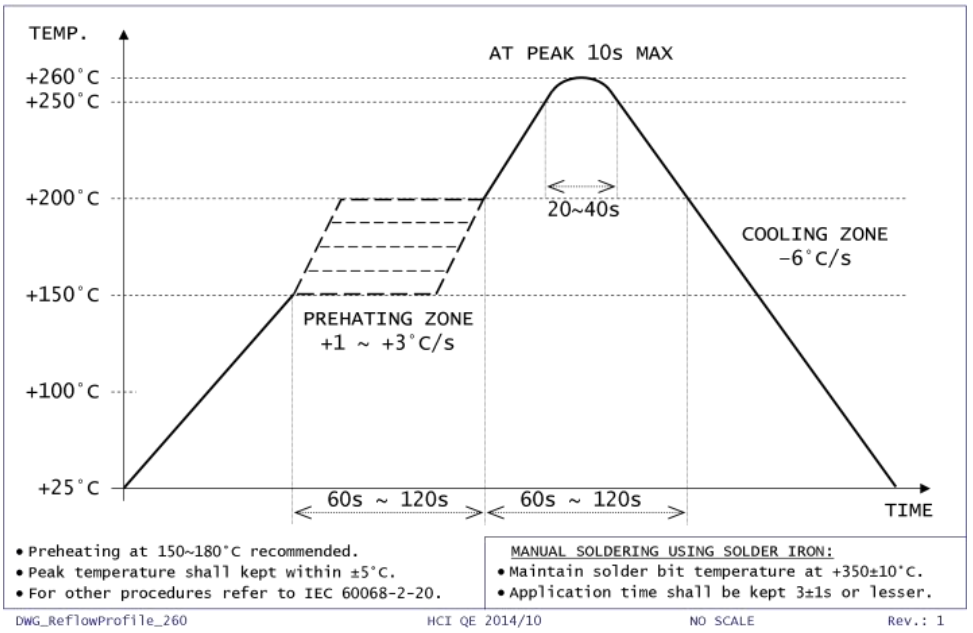
未标注公差按±0.2mm



PAD	FUNCTION
2、3、6、7	NC
1	NC
2	GND
3	Fout
4	Vcc

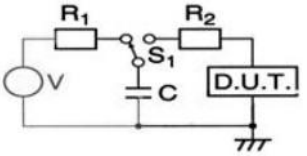
5. RECOMMENDED SOLDERING INFORMATION

RECOMMENDED REFLOW SOLDER PROFILE – PEAK TEMPERATURE UP TO +260°C



6. RELIABILITY TEST INFORMATION

	Test item	Test conditions	Criteria
1	High temperature storage	Temperature: +125°C ±5°C Time: 240 ±4 Hours Tested after 4 to 12h at room temperature.	±1.0ppm
2	Low temperature storage	Temperature: –55°C ±5°C Time: 240 ±4 Hours Tested after 4 to 12h at room temperature.	±1.0ppm
3	Temperature humidity bias THB	Temperature: +85°C ±5°C Humidity: 85% ±5% RH Time: 240 ±4 Hours BIAS: Supply Voltage Tested after 4 to 12h at room temperature.	±1.0ppm
4	Temperature cycling	Low Temp. cycle: –55°C ±2°C High Temp. cycle: +85°C ±5°C Time: 30min each cycle Number of cycles: 1,000 Tested after 4 to 12h at room temperature.	±1.0ppm
5	Aging	Temperature: +85°C ±5°C Time: 30d Tested after 4 to 12h at room temperature.	±1.0ppm
6	Resistance to solder heat	Reflow peak temp.: +260°C ±5°C (refer to rec. profile) Number of cycles: 3 times Tested after 4 to 12h at room temperature.	±1.0ppm
7	Solderability (MIL-STD-883E)	Dip in flux: 5~10 Seconds Temperature: 230℃±10℃ Time: 5 Seconds Tested after 4 to 12h at room temperature.	>95% cover.

8	Drop test	Drop height: 120cm Number of cycles: 12 times Drop height: 150cm Number of cycles: 9 times With jig (120~150g) onto iron plate Tested after 24h at room temperature.	±1.0ppm
9	Vibration	Frequency Range: 20~2000Hz PSD: 0.053g^2 Time: 40min each direction (X,Y,Z) Tested after 4 to 12h at room temperature.	±1.0ppm
10	ESD-HBM	HBM, V=±1KV, C=100pF, R1=10M, R2=1.5K, 3times 	±1.0ppm

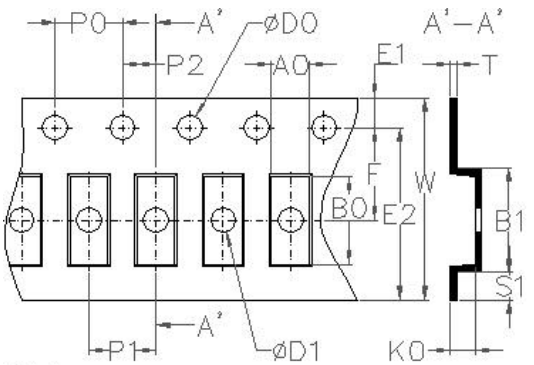
7. ENVIRONMENTAL COMPLIANCE INFORMATION

		Compliance information
1	RoHS	This product is fully RoHS compliant, 6/6 compliant per EU legislation.
2	RoHS 2	In regards of RoHS 2, CE marking directive for finished products, we can provide RoHS test reports and MDS to show compliance, but since our product is not a final application we have no CE mark.
3	Lead-Free	This product is considered Lead-Free, Lead (Pb) contamination is controlled to be below 200ppm.
4	Halogen-Free	This product is compliant to IEC 61249-2-21:2003 (Br<800ppm / Cl<800ppm).
5	REACH (SVHC)	This product does not contain substances (SVHC) listed by REACH, we continuously monitor updates of the list of SVHC's
6	PFOS / PFOA Free	This product is free of any PFOS / PFOA.
7	Electrostatic Discharge (ESD) sensitivity	This product is ESD sensitive and requires precautions for handling and storage. Follow JEITA EIAJ ED-4701 or JSD22 or ANSI-ESD-S20-20 or IEC 61000-4-2.
8	Moisture Sensitivity	This product is hermetically sealed and does NOT fall under the classification of moisture sensitivity per J-STD-020C (Standard is for non-hermetically sealed components). If required we suggest to use LEVEL 1

8. PACKAGING

Carrier

	Parameter	STANDARD PACKAGING	ALTERNATE PACKAGING
1	A0	3.6±0.1	
2	B0	5.4±0.1	
3	K0	2.4±0.1	
4	B1	6.0±0.1	
5	P0	4.0±0.1	
6	P1	4.0±0.1	
7	T	0.25±0.05	
8	W	12.0±0.2	

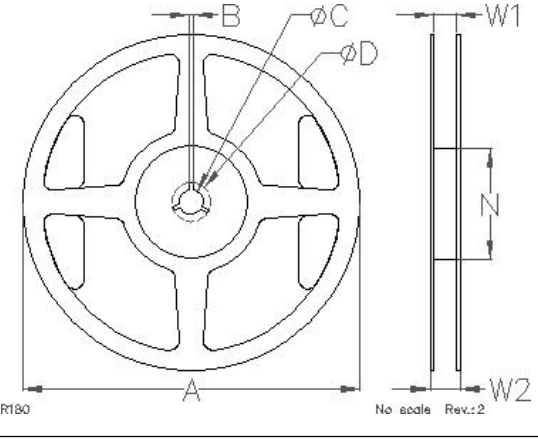


Note 1: All dimensions in [mm].
Note 2: All dimensions not specified or not being shown follow EIA-481 standard.

Reel

QTY per reel: 1,000pcs MAX

	Parameter	STANDARD PACKAGING	ALTERNATE PACKAGING
9	A	$180^{+0}_{-1.5}$	
10	B	2.0±0.5	
11	ØC	13±0.2	
12	ØD	21±0.8	
13	N	60±2	
14	W1	$12.4^{+2.0}_{-0}$	
15	W2	$16\,4^{+2.0}_{-0}$	



Note 1: All dimensions in [mm]. Dimension W1 is measured near the Hub (N).
Note 2: All dimensions not specified or not being shown follow EIA-481 standard.

Unreeling information

	Oscillator product's orientation
16	This product is a polarized component which requires a certain orientation; Pin 1 is identified on top side marking with a DOT. In the carrier tape is the component oriented with pin 1 towards the sprocket holes. (per EIA-481)

